

SIXTH SEMESTER SYLLABUS

Course Code: BCA 302T

L T C

Course Name: Distributed systems and Cloud Computing

4 0 4

INSTRUCTIONS TO PAPER SETTERS:

1. Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
2. Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
3. Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. To understand the fundamentals of distributed systems, their architecture, characteristics, and applications.
2. To learn the concepts of communication, synchronization, and resource sharing in distributed systems.
3. To understand cloud computing models, virtualization, and cloud service architectures.
4. To explore cloud platforms, cloud storage, and data-intensive computing concepts.
5. To develop an understanding of cloud-based applications and emerging trends in distributed and cloud computing.

PRE-REQUISITES:

1. Fundamentals of Computers and Information Technology.
2. Basic knowledge of Computer Networks.
3. Basic understanding of Operating Systems.

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:-

CO #	Detailed Statement of the CO
CO1	Explain the fundamental concepts, architecture, characteristics, and applications of distributed systems.
CO2	Describe communication, synchronization, resource sharing, and fault tolerance in distributed systems.
CO3	Explain cloud computing concepts, virtualization, deployment models, and service models.
CO4	Demonstrate knowledge of cloud platforms, cloud storage, and data-intensive computing techniques.
CO5	Analyze cloud-based applications and emerging trends in distributed and cloud computing.

Course Outcomes	Program Outcomes (Scale: 1 – Very Low, 2 – Low, 3 – Medium, 4 – High)									
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	2	2	2	1	2	1	1	1	1
CO2	3	3	2	3	2	2	1	1	1	1
CO3	3	2	3	3	3	2	1	1	1	2
CO4	2	3	3	3	3	2	1	2	1	2
CO5	2	3	3	3	2	3	2	2	2	3

UNIT – I

No. of Hours: 10 Chapter/Book Reference: TB1, TB2

Introduction to Distributed Systems: Definition, goals, characteristics, architecture, distributed system models, advantages and challenges.

Communication in Distributed Systems: Client-server architecture, Remote Procedure Call (RPC), Remote Method Invocation (RMI), sockets, message passing.

Applications of Distributed Systems: Distributed databases, distributed file systems, web-based applications.

UNIT – II

No. of Hours: 11 Chapter/Book Reference: TB1, TB2

Synchronization and Resource Management: Clock synchronization, mutual exclusion, process coordination, deadlocks, fault tolerance.

Introduction to Cloud Computing: Characteristics, historical evolution, cloud architecture, cloud deployment models (Public, Private, Hybrid, Community), cloud service models (IaaS, PaaS, SaaS).

UNIT – III

No. of Hours: 12 Chapter/Book Reference: TB2, TB3

Virtualization: Concepts, types of virtualizations, virtual machines, hypervisors, advantages of virtualization.

Cloud Platforms and Services: Cloud storage, cloud databases, cloud computing platforms, introduction to AWS, Microsoft Azure and Google Cloud Platform.

UNIT – IV

No. of Hours: 10 Chapter/Book Reference: TB2, TB3

Data Intensive Computing: Introduction to Big Data, MapReduce, cloud-based data processing. Cloud

Applications: E-learning, healthcare, e-commerce, social media, IoT applications. Emerging Trends: Edge computing, serverless computing, containerization, green cloud computing.

TEXT BOOKS:

TB1. George Coulouris, Jean Dollimore, Tim Kindberg and Gordon Blair, "Distributed Systems: Concepts and Design", 6th Edition, Pearson, 2021.

TB2. Rajkumar Buyya, Christian Vecchiola and S. Thamarai Selvi, "Mastering Cloud Computing: Foundations and Applications Programming", 1st Edition, Morgan Kaufmann, 2013.

TB3. Kai Hwang, Geoffrey Fox and Jack Dongarra, "Distributed and Cloud Computing: From Parallel Processing to the Internet of Things", 1st Edition, Morgan Kaufmann, 2012.

REFERENCE BOOKS:

RB1. Andrew S. Tanenbaum and Maarten Van Steen, "Distributed Systems", 4th Edition, Pearson, 2024.

RB2. Thomas Erl, Ricardo Puttini and Zaigham Mahmood, "Cloud Computing: Concepts, Technology & Architecture", Pearson, 2013.

Applicable from Batch Admitted in Academic Session 2024-25 onwards

RB3. Barrie Sosinsky, "Cloud Computing Bible", 2nd Edition, Wiley, 2019.

RB4. Michael J. Fischer, Nancy Lynch and Michael S. Paterson, "Distributed Computing: Principles, Algorithms and Systems", latest available edition

Course Code: BCA 304T

L T C

Course Name: Machine Learning

4 0 4

INSTRUCTIONS TO PAPER SETTERS:

1. Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
2. Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
3. Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. To make student able to learn mathematical concepts, and algorithms used in machine learning techniques for solving real world problems and developing new applications based on machine learning.
2. To introduce students to the state-of-the-art concepts and techniques of Machine Learning using Python.

PRE-REQUISITES:

1. Basics of Python Programming.

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:-

CO #	Detailed Statement of the CO
CO1	Explain machine learning concepts on real world applications and problems
CO2	Analyze and Implement Regression techniques.
CO3	Solve and design solution of Classification problem.
CO4	Understand and implement Unsupervised learning algorithms.
CO5	Interpret various machine learning algorithms in a range of real-world applications

Course Outcomes	Program Outcomes (Scale: 1 – Very Low, 2 – Low, 3 – Medium, 4 – High)									
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	4	3	3	3	2	4	1	2	1	2
CO2	4	2	4	4	2	4	1	3	1	2
CO3	4	2	4	4	2	4	1	3	1	2

Applicable from Batch Admitted in Academic Session 2024-25 onwards

CO4	4	2	4	4	2	4	1	3	1	2
CO5	4	3	4	4	3	4	2	4	2	3

UNIT – I

No. of Hours: 11 **Chapter/Book Reference: TB1, TB2**

Introduction to Machine Learning, Why Machine learning, Types of Machine Learning Problems, Applications of Machine Learning. Supervised Machine Learning- Regression and Classification. Binary Classifier, Multiclass Classification, Multilabel Classification. Performance Measures- Confusion Matrix, Accuracy, Precision & recall, ROCCurve. Advanced Python- NumPy, Pandas. Python Machine Learning Library Scikit-Learn, Linear Regression with one Variable, Linear Regression with Multiple Variables, Logistic Regression.

UNIT – II

No. of Hours: 11 **Chapter/Book Reference: TB1, TB2**

Supervised learning Algorithms: Decision Trees, Tree pruning, Rule-based Classification, Naïve Bayes, Bayesian Network. Support Vector Machines, k-Nearest Neighbor, Ensemble Learning and Random Forest algorithm.

UNIT – III

No. of Hours: 11 **Chapter/Book Reference: TB1, TB3**

Artificial Neural Networks, HebbNet, Perceptron, Adaline, Multilayer Neural Network, Architecture, Activation Functions, Loss Function, Hyper parameters, Gradient Descent, Backpropagation, Variants of Backpropagation, Avoiding overfitting through Regularization, Applications of Neural Networks.

UNIT – IV

No. of Hours: 11 **Chapter/Book Reference: TB1, TB2, TB3**

Unsupervised learning algorithms: Introduction to Clustering, K-means Clustering, Hierarchical Clustering, Kohonen Self-Organizing Maps. Implementation of Unsupervised algorithms. Feature selection and Dimensionality reduction, Principal Component Analysis.

TEXT BOOKS:

TB1. Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow", 3rd Edition, O'Reilly Media, 2022.

TB2. U. Dinesh Kumar and Manaranjan Pradhan, "Machine Learning using Python", Wiley, 2019.

TB3. Laurence Fausett, "Fundamentals of Neural Networks: Architectures, Algorithms and Applications", Pearson Education, 1st Edition, 1994

REFERENCE BOOKS:

RB1. Tom M. Mitchell, "Machine Learning", 1st Edition, McGraw-Hill Education, 1997.

RB2. Timothy A. Budd, "Exploring Python", 1st Edition, McGraw-Hill Education, 2011.

RB3. Jake VanderPlas, "Python Data Science Handbook: Essential Tools for Working with Data", 1st Edition, O'Reilly Media, 2016

Course Code: BCA 306T

L T C

Course Name: Software Project Management

4 0 4

INSTRUCTIONS TO PAPER SETTERS:

1. Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
2. Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
3. Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced way.

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. To understand the fundamentals of software project management and software development life cycle.
2. To learn project planning, estimation and scheduling techniques used in software projects.
3. To understand project execution, quality assurance and risk management practices.
4. To develop effective communication and teamwork skills required for software project management.
5. To understand project monitoring, stakeholder management and software project documentation.

PRE-REQUISITES:

1. Fundamentals of Software Engineering.

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:-

CO #	Detailed Statement of the CO
CO1	Explain the fundamental concepts, phases and activities involved in software project management.
CO2	Apply project planning, estimation and scheduling techniques for software projects.
CO3	Describe project execution, quality assurance, risk management and configuration management practices.
CO4	Demonstrate effective communication, teamwork and stakeholder management in software projects.
CO5	Analyze software project progress using project monitoring, reporting and documentation techniques.

Course Outcomes	Program Outcomes (Scale: 1 – Very Low, 2 – Low, 3 – Medium, 4 – High)									
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	2	2	2	2	1	1	2	1	1
CO2	3	3	3	2	3	1	1	2	1	2
CO3	3	3	3	3	3	2	1	2	1	2
CO4	2	2	2	2	2	2	1	3	2	2
CO5	3	3	3	3	3	2	2	2	2	3

UNIT – I

No. of Hours: 10 Chapter/Book Reference: TB1, TB2, TB3

Introduction to Software Project Management: Project management concepts, software project characteristics, project life cycle, Software Development Life Cycle (SDLC), role and responsibilities of the project manager, project stakeholders, project constraints, project feasibility study.

UNIT – II

No. of Hours: 11 Chapter/Book Reference: TB1, TB2

Project Planning and Estimation: Project initiation, project scope, Work Breakdown Structure (WBS), project estimation techniques (COCOMO and Function Point Analysis), scheduling using Gantt Charts and PERT, resource allocation, project budgeting, project documentation, project milestones and deliverables.

UNIT – III

No. of Hours: 11 Chapter/Book Reference: TB1, TB2, TB3

Project Execution and Control: Software development methodologies (Waterfall and Agile), team organization and roles, project monitoring and control, software quality assurance, software testing basics, risk management, change management, configuration management, project performance evaluation.

UNIT – IV

No. of Hours: 10 Chapter/Book Reference: TB1, TB2, TB3

Project Communication and Maintenance: Communication planning, stakeholder identification and management, team collaboration, project reporting, status meetings, software maintenance, project closure, software project documentation, customer relationship management, case studies of software projects.

TEXT BOOKS:

TB1. Kathy Schwalbe, "Information Technology Project Management", 10th Edition, Cengage Learning, 2024.

TB2. Bob Hughes, Mike Cotterell and Rajib Mall, "Software Project Management", 7th Edition, McGraw Hill Education, 2022.

TB3. Roger S. Pressman and Bruce R. Maxim, "Software Engineering: A Practitioner's Approach", 9th Edition, McGraw Hill Education, 2023.

REFERENCE BOOKS:

RB1. Kim Heldman, "Project Management JumpStart", 4th Edition, Wiley, 2018.

RB2. Scott Berkun, "Making Things Happen: Mastering Project Management", Revised Edition, Applicable from Batch Admitted in Academic Session 2024-25 onwards

O'Reilly Media, 2008.

RB3. Andrew Stellman and Jennifer Greene, "Head First PMP", 4th Edition, O'Reilly Media, 2021.

RB4. Timothy C. Lister and Tom DeMarco, "Peopleware: Productive Projects and Teams", 3rd Edition, Addison-Wesley Professional, 2013

Course Code: BCA 312T

L T C

Course Name: Data Visualization & Analytics

4 0 4

INSTRUCTIONS TO PAPER SETTERS:

1. Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
2. Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
3. Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. Students will develop relevant programming abilities.
2. Students will demonstrate proficiency with statistical analysis of data.
3. Conduct exploratory data analysis using visualization.
4. Craft visual presentations of data for effective communication.

PRE-REQUISITES:

1. Basics of Python Programming.

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:-

CO #	Detailed Statement of the CO
CO1	Illustrating the features of Multithreading in python
CO2	Analyzing data using suitable python library.
CO3	Visualizing data using Matplotlib, Seaborn library.
CO4	Develop python applications with database connectivity operations.

Course Outcomes	Program Outcomes (Scale: 1 – Very Low, 2 – Low, 3 – Medium, 4 – High)									
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	4	2	4	3	2	4	1	2	1	1
CO2	4	2	4	4	2	4	1	3	1	2
CO3	4	2	4	4	3	4	1	3	2	2
CO4	4	2	4	4	4	4	2	3	2	2

UNIT – I

No. of Hours: 11 **Chapter/Book Reference: TB1, TB2**

Analytics: Basic Nomenclature, Analytics Process Model, Analytics part in different profiles, Analytical
Applicable from Batch Admitted in Academic Session 2024-25 onwards

Model Requirements.

Data Sources for data collection, Sampling and Sampling distribution, Types of data elements, Missing Values, Outlier Detection and Treatment, Standardization using Min/max and z-score, categorization, Segmentation.

UNIT – II

No. of Hours: 11 **Chapter/Book Reference: TB2, TB3**

Statistical Hypothesis Testing, p-Values, Confidence Intervals.

Correlation, Simpson's Paradox, Some Other Correlational Caveats, Correlation and Causation, Correlation Statistics-ANOVA.

UNIT – III

No. of Hours: 11 **Chapter/Book Reference: TB3, TB4**

Data Visualization: Graphs in Python: Line Graph, Bar charts, Pie-charts, Scatter plots, multiple plots, Subplots, Legends, changing figure Size, Styling plots using Matplotlib Library. Functions like relplot(), displot() and catplot (). Seaborn Library: Introduction, Line plot, Distplot, Lmplot, Count plot, Color palettes.

UNIT – IV

No. of Hours: 11 **Chapter/Book Reference: TB5, TB3**

GUI Programming: Creating User-interface, GUI Widgets with Tkinter, Creating Layouts, Check Box, Radio Buttons, List Box, Menus, Menus Options, Dialog Boxes

Database Access: Database Connectivity Operations: Create, Insert, Select, Delete, Drop, Update.

TEXT BOOKS:

TB1. Bart Baesens, "Analytics in a Big Data World: The Essential Guide to Data Science and its Applications", Wiley, 1st Edition, 2014.

TB2. UEMC Education Services, "Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data", Wiley, 1st Edition, 2015.

TB3. Joel Grus, "Data Science from Scratch: First Principles with Python", O'Reilly Media, 2nd Edition, 2019.

TB4. Wes McKinney, "Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter", O'Reilly Media, 3rd Edition, 2022.

TB5. John E. Grayson, "Python and Tkinter Programming", Manning Publications, 2000.

REFERENCE BOOKS:

RB1. Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", O'Reilly Media, 2nd Edition, 2015.

RB2. Reema Thareja, "Python Programming using Problem Solving Approach", Oxford University Press, 1st Edition, 2017.

RB3. Tony Gaddis, "Starting Out with Python", Pearson, 5th Edition, 2024.

RB4. Y. Daniel Liang, "Introduction to Programming Using Python", Pearson, 1st Edition, 2013.

RB5. Mark Summerfield, "Programming in Python 3: A Complete Introduction to the Python Language", Addison-Wesley Professional, 2nd Edition, 2010.

RB6. Joel Grus, "Data Science from Scratch: First Principles with Python", O'Reilly Media, 2nd Edition, 2019.

Course Code: BCA 314T

L T C

Course Name: Deep Learning with Python

4 0 4

INSTRUCTIONS TO PAPER SETTERS:

1. Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
2. Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
3. Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. To present the mathematical, statistical and computational challenges of building neural networks
2. To study the concepts of deep learning and important deep learning techniques
3. To introduce important Deep Learning architectures
4. To enable the students to apply deep learning techniques to support real-life applications.

PRE-REQUISITES:

1. Basics of Python Programming.

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:-

CO #	Detailed Statement of the CO
CO1	Understand the basic concepts of Deep Learning and differentiate between shallow learning and deep learning
CO2	Implement various Deep Learning Models.
CO3	Understand different Deep Learning architectures and training algorithms
CO4	Understanding Dimensionality Reduction and optimization in Deep Learning.
CO5	Understanding and implementing Recurrent Neural Network(RNN).27
CO6	Applying Deep Learning techniques in real life applications such as object detection and analysis

Course Outcomes	Program Outcomes (Scale: 1 – Very Low, 2 – Low, 3 – Medium, 4 – High)									
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	4	2	3	3	2	4	1	2	1	2
CO2	4	2	4	4	3	4	1	3	1	2
CO3	4	2	4	4	3	4	1	3	1	2

Applicable from Batch Admitted in Academic Session 2024-25 onwards

CO4	4	2	4	4	2	4	1	4	1	2
CO5	4	2	4	4	3	4	1	3	1	2
CO6	4	3	4	4	3	4	1	4	2	3

UNIT – I

No. of Hours: 11 **Chapter/Book Reference: TB1, TB2**

Introduction – Overview of Machine Learning, Introduction to Artificial Neural Network (ANN), Perceptron, training a Neural Network, Activation Functions, Loss Function, Hyperparameters, Gradient Descent, Stochastic Gradient Descent, Backpropagation and regularization, Batch normalization, Building an ANN in Python, Frameworks-TensorFlow, Keras.

UNIT – II

No. of Hours: 12 **Chapter/Book Reference: TB1, TB2**

What is Deep Learning? Deep vs Shallow Networks, Convolution Neural Networks (CNN) –Convolution Layers, Pooling Layer, Flattening, Fully Connected Layers, Softmax and Cross-Entropy, building a CNN in Python, Fully Connected CNN, CNN Architectures – LeNet, AlexNet, ZFNet, GoogLeNet, VGGNet, ResNet,DenseNet, Training a Convnet: weights initialization, batch normalization, hyperparameter optimization.

UNIT – III

No. of Hours: 11 **Chapter/Book Reference: TB1**

Deep Belief Networks, Auto Encoders, Concept of Dimensionality Reduction, Autoencoder, Denoising Autoencoders, Deep Autoencoders, Concept of Reinforcement Learning.

UNIT – IV

No. of Hours: 11 **Chapter/Book Reference: TB2, TB3**

Recurrent Neural Networks (RNN), LSTM, Sequence Prediction and Time Series Forecasting with LSTM, Overview of Object Detection Techniques using Deep Learning, Overview of Transfer Learning.

TEXT BOOKS:

TB1. Adam Gibson and Josh Patterson, Deep Learning: A Practitioner's Approach, O'Reilly Media, First Edition, 2017.

TB2. Mohamed Elgendy, Deep Learning for Vision Systems, Manning Publications, First Edition, 2020.

TB3. Navin Kumar Manaswi, Deep Learning with Applications Using Python, Apress, First Edition, 2018.

REFERENCE BOOKS:

RB1. Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, Cambridge University Press, First Edition, 2015.

RB2. Li Deng and Dong Yu, Deep Learning: Methods and Applications, Now Publishers Inc., First Edition, 2014.

RB3. Ian Goodfellow, Yoshua Bengio, Aaron Courville, Deep Learning, MIT Press, First Edition, 2016.

RB4. Michael Nielsen, Neural Networks and Deep Learning, Determination Press, First Edition, 2015

RB5. Charu C. Aggarwal, Neural Networks and Deep Learning, Springer, First Edition, 2018.

RB6. M. Arif Wani, Farooq Ahmad Bhat, Saduf Afzal, Asif Iqbal Khan, Advances in Deep Learning, Springer, First Edition, 2020

Course Code: BCA 316T

L T C

Course Name: Web Security

4 0 4

INSTRUCTIONS TO PAPER SETTERS:

4. Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
5. Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
6. Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

6. To understand and learn web application and its Architecture.
7. Students will be able to develop awareness regarding Cyber laws and crimes.
8. Students will be able to understand the internet and web application security issues.
9. Students will be able to learn and understand wireless network security issues.
10. To learn and understand the concept of web services, ajax and other technology which are helpful.

PRE-REQUISITES:

4. Computer Network
5. C/C++/HTML (Programming Knowledge of C/C++/HTML/JS)

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:-

CO #	Detailed Statement of the CO
CO1	Define overall web security infrastructure, components, issues and basic concept etc.
CO2	Describe briefly various types of security like social media security, email security, web application and web services security etc. Explain Web related services.
CO3	Apply and implementing various vulnerabilities for Ethically hacking a websites / Web Applications.
CO4	Focusing Penetration Testing, Computer Forensics.
CO5	Evaluate different web security algorithms with the help of program.
CO5	Design and implement XSS attacks, SQL Injection attack, password hashing and cracking

Course Outcomes	Program Outcomes (Scale: 1 – Very Low, 2 – Low, 3 – Medium, 4 – High)									
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	3	2	2	2	2	1	1	1	1	1
CO2	3	3	2	2	3	1	1	2	1	2
CO3	3	3	3	3	3	2	1	2	1	2
CO4	3	3	3	3	2	2	1	2	2	2
CO5	3	3	3	3	3	2	2	2	2	3

UNIT – I

No. of Hours: 12 Chapter/Book Reference: TB1, TB2

Components of Internet, Weak points of Internet, HTTP vs HTTPS, Overview of web authentication technologies, Web application architecture, Recent attack trends, Types of Web Security, Web infrastructure security/Web application firewalls, managing configurations for web apps, Techniques of Web Hacking, Methods of Attacking users, Importance of Web Application Security, Web Application Security vs Network Security. Social Media security - What is Online Social Networks, data collection from social networks, challenges, opportunities, and pitfalls in online social networks, APIs Collecting data from Online social media. Trust, credibility, and reputations in social systems.

UNIT-II

No. of Hours: 11 Chapter/Book Reference: TB1, TB2

Internet and Web Application Security: Email security (PGP and SMIME), Web Security: Web authentication, Injection Flaws, Programming Bugs and Malicious code, XSS and SQL Injection, Memory corruption exploits, Web Browser Security, E-Commerce Security

UNIT-III

No. of Hours: 11 Chapter/Book Reference: TB1, TB2

Wireless Network Security: Components, Security issues, Securing a Wireless Network, Mobile Security Management: Disaster Recovery, Ethical Hacking, Penetration Testing, Computer Forensics, Cyber laws and crime, Security Audit and Investigation, Cyber Security Solutions

UNIT-IV

No. of Hours: 10 Chapter/Book Reference: TB1, TB2

Web services overview, Honeypot, XML security, AJAX attack trends and common attacks, REST security, Content Security Policy Serialization security, Clickjacking, DNS rebinding, HTML5 security, Logging collection and analysis for web apps, Security testing, IPv6 impact on web security

TEXT BOOKS:

TB1. Joel Scambray, Mike Shema and Caleb Sima, "Hacking Exposed Web Applications: Web Application Security Secrets and Solutions", 3rd Edition, McGraw-Hill Education, 2010. (Latest edition available.)

TB2. Rafay Baloch, "Ethical Hacking and Penetration Testing Guide", CRC Press, 2015.

TB3. Georgia Weidman, "Penetration Testing: A Hands-On Introduction to Hacking", 2nd Edition, No Starch Press, 2021.

REFERENCE BOOKS:

RB1. Dafydd Stuttard and Marcus Pinto, "The Web Application Hacker's Handbook: Finding and Exploiting Security Flaws", 2nd Edition, Wiley, 2011.

RB2. EC-Council, "Computer Forensics: Investigating Network Intrusions and Cyber Crime", 2nd Edition, Cengage Learning, 2010.

Applicable from Batch Admitted in Academic Session 2024-25 onwards

- RB3.** John W. Rittinghouse and William M. Hancock, "Cyber Security Operations Handbook", Elsevier, 2020.
- RB4.** Deborah G. Johnson, "Computer Ethics", 4th Edition, Pearson Education, 2009.
- RB5.** Ernest A. Kallman and John P. Grillo, "Ethical Decision Making and Information Technology: An Introduction with Cases", 6th Edition, McGraw-Hill Education, 2016.

Course Code: BCA 318T

L T C

Course Name: Mobile Application Development

4 0 4

INSTRUCTIONS TO PAPER SETTERS:

1. Question No. 1 should be compulsory and cover the entire syllabus. There should be 10 questions of short answer type of 2 marks each, having at least 2 questions from each unit.
2. Apart from Question No. 1, rest of the paper shall consist of four units as per the syllabus. Every unit should have two questions to evaluate analytical/technical skills of candidate. However, student may be asked to attempt only 1 question from each unit. Each question should be of 10 marks, including its subparts, if any.
3. Examiners are requested to go through the Course Outcomes (CO) of this course and prepare the question paper accordingly, in such a way that every question be mapped to some or other CO and all the questions, put together, must be able to achieve the mapping to all the CO(s), in balanced

LEARNING OBJECTIVES:

In this course, the learners will be able to develop expertise related to the following:

1. Identify various concepts of mobile application programming that make it unique from programming for other platforms.
2. To help learner to gain a basic understanding of Android application development.
3. Program mobile applications for the Android operating system that use basic and advanced phone features, and deploy applications to the Android marketplace for distribution.

PRE-REQUISITES:

1. Java Programming.
2. Programming fundamental

COURSE OUTCOMES (COs):

After completion of this course, the learners will be able to:-

CO #	Detailed Statement of the CO
CO1	Recognize the concept of application development for mobile devices.
CO2	Understand the basic technologies used by the Android platform.
CO3	Recognize and use Android Environment Emulator and Application life cycle
CO4	Develop mobile applications for the Android operating system that use basic and advance phone features
CO5	Deploy applications to the Android marketplace for distribution

Course Outcomes	Program Outcomes (Scale: 1 – Very Low, 2 – Low, 3 – Medium, 4 – High)									
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	4	2	3	2	2	4	1	1	1	2
CO2	4	2	3	3	3	4	1	2	1	2
CO3	4	2	4	4	3	4	1	2	2	2

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CO4	4	2	4	4	4	4	2	3	2	2
CO5	4	3	4	4	4	4	2	3	2	3

UNIT – I

No. of Hours: 11 **Chapter/Book Reference: TB1, TB2**

Introduction: Brief history of mobile applications, Different types of mobile applications, Brief history of Android, Introduction to Android Development Environment, Android Application Design Essentials: Anatomy of an Android applications, Creating First Android Application, Creating Android project , Project organization, Setting up real Android device , Setting up Android emulator, Developing simple user interface , Running your first application Android terminologies, Application Context, Activities, Services, Intents, Receiving and Broadcasting Intents, Android Manifest File and its common settings, Using Intent Filter, Permissions.

UNIT – II

No. of Hours: 11 **Chapter/Book Reference: TB1, TB2**

User Interface in Android: Adaptive and responsive user interfaces, User Input Controls, Menus, Screen Navigation, RecyclerView, Drawables, Themes and Styles, Fragments Fragment Life Cycle, Introduction to Material Design. Android Application Components: App Widgets, Processes and Threads, User Interface Components, Views and layouts, Input controls, Input Events, Settings, Dialogs, Menus, Notifications, Toasts, Testing the user interface.

UNIT – III

No. of Hours: 11 **Chapter/Book Reference: TB1, TB2**

Background tasks: AsyncTask, AsyncTask Loader, Connecting App to Internet, Broadcast receivers, Services, Notifications, Alarm managers. Sensor, Location and Maps: Sensor Basic, Motion and Position Sensors, Location services, Google maps API, Google Places API.

UNIT – IV

No. of Hours: 11 **Chapter/Book Reference: TB1, TB2**

Working with data in Android: Shared Preferences, App Setting, SQLite primer, Store data using SQLite database, Content Providers, Content Resolver, Loader Publishing Your App: Preparing for publishing, Signing and preparing the graphics, publishing to the Android Market Using Common Android APIs: Using Android Data and Storage APIs, Managing data using Sqlite, Sharing Data between Applications with Content Providers, Using Android Networking APIs, Using android Web APIs, Using Android Telephony APIs, Deploying Android Application to the World.

TEXT BOOKS:

TB1. Lauren Darcey and Shane Conder, Android Wireless Application Development, Pearson Education.

TB2. Wei-Meng Lee, Beginning Android Application Development: Android 6 Edition, Wiley India Pvt. Ltd., 2015.

TB3. J. F. DiMarzio, Android: A Programmer's Guide, McGraw-Hill Education (India) Private Limited, 2008

REFERENCE BOOKS:

RB1. aul Deitel, Harvey Deitel, Abbey Deitel and Michael Morgano, Android for Programmers: An App-Driven Approach, Pearson Education, 2015.

RB2. Wei-Meng Lee, Beginning Android Application Development, Wiley Publishing, 2015.

Course Code: BCA 320
Course Name: Minor Project

L T/P C
0 14 6

PROJECT REPORT

All the students are required to submit a report based on the project work done by them during the sixth semester.

SYNOPSIS (SUMMARY/ABSTRACT) :

All students must submit a summary/abstract separately with the project report. Summary, preferably, should be of about 3-4 pages. The content should be as brief as is sufficient enough to explain the objective and implementation of the project that the candidate is going to take up. The write up must adhere to the guidelines and should include the following:

- Name / Title of the Project
- Statement about the Problem
- Why is the particular topic chosen?
- Objective and scope of the Project
- Methodology (including a summary of the project)
- Hardware & Software to be used
- Testing Technologies used
- What contribution would the project make?

TOPIC OF THE PROJECT- This should be explicitly mentioned at the beginning of the Synopsis. Since the topic itself gives a peep into the project to be taken up, candidate is advised to be prudent on naming the project. This being the overall impression on the future work, the topic should corroborate the work.

OBJECTIVE AND SCOPE: This should give a clear picture of the project. Objective should be clearly specified. What the project ends up to and in what way this is going to help the end user has to be mentioned.

PROCESS DISCRIPTION: The process of the whole software system proposed, to be developed, should be mentioned in brief. This may be supported by DFDs / Flowcharts to explain the flow of the information.

RESOURCES AND LIMITATIONS: The requirement of the resources for designing and developing the proposed system must be given. The resources might be in form of the hardware/software or the data from the industry. The limitation of the proposed system in respect of a larger and comprehensive system must be given.

CONCLUSION: The write-up must end with the concluding remarks- briefly describing innovation in the approach for implementing the Project, main achievements and also any other important feature

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that makes the system stand out from the rest.

The following suggested guidelines must be followed in preparing the Final Project Report:

Good quality white A4 size paper should be used for typing and duplication. Care should be taken to avoid smudging while duplicating the copies.

Page Specification: (Written paper and source code)

- Left margin - 3.0 cms
- Right margin- 2.0cms
- Top margin 2.54cms
- Bottom margin 2.54cms
- Page numbers - All text pages as well as Program source code listing should be numbered at the bottom center of the pages.

Normal Body Text: Font Size: 12, Times New Roman, Double Spacing, Justified. 6 point above and below para spacing

Paragraph Heading Font Size: 14, Times New Roman, Underlined, Left Aligned. 12 point above & below spacing.

Chapter Heading Font Size: 20, Times New Roman, Centre Aligned, 30 point above and below spacing.

Coding Font size :10, Courier New, Normal.

Submission of Project Report to the University: The student will submit his/her project report in the prescribed format.

The Project Report should include:

1. One copy of the summary/abstract.
2. One hard Copy of the Project Report.
3. The Project Report may be about 75 pages (excluding coding).

FORMAT OF THE STUDENT PROJECT REPORT ON COMPLETION OF THE PROJECT

I. Cover Page as per format

II. Acknowledgement

III. Certificate of the project guide

IV. Synopsis of the Project

V. Main Report

i. Objective & Scope of the Project

ii. Theoretical Background Definition of Problem

iii. System Analysis & Design vis-a-vis User Requirements

iv. System Planning (PERT Chart)

v. Methodology adopted, System Implementation & Details of Hardware & Software used System Maintenance & Evaluation

vi. Detailed Life Cycle of the Project

a. ERD, DFD

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- b. Input and Output ScreenDesign
- c. Processinvolved
- d. Methodology usedtesting
- e. Test Report, Printout of the Report & CodeSheet
- VI. Coding and Screenshots of the project
- VII. Conclusion and Future Scope
- VIII. References

Formats of various certificates and formatting styles are as:

1. Certificate from the Guide
CERTIFICATE This is to certify that this project entitled “ xxxxxx
xxxxxx xxxxxx xxxxx xxxxx xxx” submitted in partial fulfillment of the degree of Bachelor of
Computer Applications to the “xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx” through xxxxxx xxxxxx
doneby Mr./Ms.,RollNo..... is an is an authentic
work carried out by him/heratunder my guidance.
The matter embodied in this project work has not been submitted earlier for award of any degree
to the best of my knowledge andbelief.

Signature of the student

Signature of the Guide

2. Project Report Cover PageFormat:

Title of the Project/report
(Times New Roman, Italic, Font size = 24)

**Submitted in partial fulfilment of the requirements for the award of the
degree of**

Bachelor of Computer Applications
(Bookman Old Style, 16 point, centre)

Submitted to:
(Guide Name)

Submitted by:
(Student'sname)
Roll No
College Name

3. Self-Certificate by the students

SELF CERTIFICATE

This is to certify that the dissertation/project report entitled “.....” is
done by me is an authentic work carried out for the partial fulfilment of the requirements for the
Applicable from Batch Admitted in Academic Session 2024-25 onwards

award of the degree of Bachelor of Computer Applications under the guidance of.....
The matter embodied in this project work has not been submitted earlier for award of any degree or diploma to the best of my knowledge and belief.

Signature of the student

Name of the Student

Roll No.

4. ACKNOWLEDGEMENTS

In the “Acknowledgements” page, the writer recognizes his indebtedness for guidance and assistance of the thesis adviser and other members of the faculty. Courtesy demands that he also recognizes specific contributions by other persons or institutions such as libraries and research foundations. Acknowledgements should be expressed simply, tastefully, and tactfully.

Course Code: BCA 302P **L P C**
Course Name: Distributed systems and Cloud Computing Lab **0 2 1**

INSTRUCTIONS

1. The course objectives and course outcomes are identical to that of BCA 302T as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class

Course Code: BCA 304P **L P C**
Course Name: Machine Learning Lab **0 2 1**

INSTRUCTIONS

1. The course objectives and course outcomes are identical to that of BCA 304T as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class

Course Code: BCA 306P **L P C**
Course Name: Software Project Management Lab **0 2 1**

INSTRUCTIONS

1. The course objectives and course outcomes are identical to that of BCA 306T as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class

Course Code: BCA 312P **L P C**
Course Name: Data Visualization & Analytics Lab **- 4 2**

INSTRUCTIONS

1. The course objectives and course outcomes are identical to that of BCA 312T as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class

Course Code: BCA 314P **L P C**
Course Name: Deep Learning with Python Lab **- 4 2**

INSTRUCTIONS

1. The course objectives and course outcomes are identical to that of BCA 314T as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement

Course Code: BCA 316P
Course Name: Web Security Lab

L P C
- 4 2

INSTRUCTIONS

1. The course objectives and course outcomes are identical to that of BCA 316T as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement

Course Code: BCA 318P
Course Name: Mobile Application Development Lab

L P C
- 4 2

INSTRUCTIONS

1. The course objectives and course outcomes are identical to that of BCA 318T as this is the practical component of the corresponding theory paper.
2. The practical list shall be notified by the teacher in the first week of the class commencement